

An adaptive time-graded control method for VSC-MTDC networks

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ABSTRACT

Power flow and voltage control, beside other challenges such as DC faults, are the main concerns in the operation of multi-terminal HVDC (MTDC) networks. This paper proposes a novel DC voltage control method, which is inspired from the principles of power system protection, for VSC-based MTDC networks. This time-graded control method periodically ranks all converters; the highest priority one is dedicated to regulate the network voltage while the others operate in the power control mode. In abnormal conditions in which the grid-forming converter cannot solely control the network voltage, a sufficient number of converters are sequentially turned to the voltage control mode according to their priority. Upon recovering from the abnormal condition, the proposed controller with its conditional reclosing ability returns these mode-changed converters to the power control mode. Since the proposed method regulates the voltage of MTDC network by the minimum number of converters, the maximum ones can be employed in power flow control. Proposing an adaptive procedure to calculate appropriate mode change delays is another contribution of this communication-less control method. Simulation of a test network in MATLAB/Simulink under various conditions with both proposed and conventional droop controllers verifies the outstanding abilities of the proposed method.

1. Introduction

HVDC system has several advantages over conventional HVAC transmission networks. Flexibility and the ability of transmitting higher amount of power over long distances with lower power losses are some of these advantages to name a few. These make the HVDC system attractive for both industries and academics [1,2]. Besides, it has been considered as an appropriate solution for some applications such as integration of remote renewable power generation sites [3–5], delivering bulk power over long distances [6] and interconnecting asynchronous AC systems [7].

According to the continuing efforts for building a HVDC network with appropriate characteristics, two different converter technologies have been introduced. These include thyristor-based or the line commutated converter (LCC), and the voltage source converter (VSC) [8]. Comparing with the former, VSC-based HVDC system presents distinctive benefits such as independent control of the active and reactive powers, less environment impacts, no commutation failure and no need to reactive compensators [9]. Besides, a multi-terminal HVDC system (MTDC), as the future of HVDC networks, can be easily established by VSCs.

The MTDC network, as a promising concept, has recently motivated a broad group of researchers in order to make it more applicable in

modern power systems. For example, CIGRÉ working groups B4-56 to B4-60 are trying to standardize MTDC networks [10]. Moreover, some large scale projects have been carried out and several others are also under construction around the world [11,12]. Since the MTDC network affects its adjacent AC grids, transmission system operators (TSOs) should specify requirements for integration of MTDC into the conventional HVAC systems [13].

In HVAC systems, both amplitude and phase of voltages are employed to control the voltage as well as power flow of network. Since the voltage and power of each converter cannot be simultaneously controlled solely with the amplitude of its DC voltage, the operation of MTDC networks is more challenging [14]. Considering this concern, different DC voltage control methods with their benefits and drawbacks have been introduced [15].

The droop control method, emanated from conventional HVAC systems, is widely utilized and preferred for microgrids as well as renewable energy systems [16,17]. In this method, the active power exchange of each converter is directly proportional to the local deviation of its DC side voltage. By droop control method, each power flow variation can be quickly responded with no control mode change or need to a remote communication link. However, deviation of DC voltages from their rated values, frequency deviation in the connected AC networks, converter overloading and complexity of power flow

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calculations are the consequences of this control method [18]. Therefore, some developed droop controllers have been introduced to alleviate these problems [19–22]. Other well-known DC voltage control methods for the VSC-MTDC network, which are described in the Section 2, include ratio, master-slave, priority, and voltage margin control [15].

In this paper, an adaptive time-graded DC voltage control method is proposed for VSC-MTDC networks. At first, all converters are periodically ranked at the base of their ability to regulate network voltage. In the next step, the highest priority one, as the slack converter, controls the network voltage while the others operate in the power control mode. Abnormal conditions such as converter outage, DC fault and large change in the power set-point lead to inability of the slack converter to regulate the network voltage. Consequently, large voltage variations, as the indicator of intolerable power balance, occur in the MTDC network. In these conditions, the minimum number of high priority converters will be turned to the voltage control mode, after their adaptive time delay. Thus, the network voltage can be regulated under various conditions by the minimum number of converters, however power flow of the MTDC network is still controlled by the others.

This paper is organized as follows; the well-known DC voltage control methods of VSC-MTDC networks are briefly described in Section 2. Those concepts of power system protection, which are required for description of the adaptive time-graded control method, are explained in Section 3. Thereafter, the proposed control method is described in details. The Section 5 is dedicated to validation of the proposed control method under various conditions. Simulation of the test system with the proposed as well as droop controllers by MATLAB/Simulink verify the outstanding advantages of the proposed DC voltage control method.

2. DC voltage control methods for VSC-MTDC networks

In order to control the power flow of VSC-MTDC network, a multi-level hierarchical control structure is usually employed [23]. In the lowest level, which is commanded by the secondary control one, the injected current of each VSC is controlled to regulate its terminal voltage. The tertiary level, as the highest level of hierarchical control structure, is responsible for dispatching power among VSCs to obtain some specific goals such as power losses minimization [15]. This supervisory control system ensures power balance, prevents violation of voltage as well as current limits after contingencies and provides an auxiliary service to adjacent AC systems [24]. Despite the participation of these control levels in the power flow control, the present study proposes a novel method for the secondary control level. Among various introduced methods for this level such as distributed control strategy [5], five well-known ones are briefly described in the following.

2.1. Droop control

As it was explained in the previous section, the advantages of droop control make this method as the most practical one for MTDC networks. However, changing the equilibrium point of the system in consequence of each new loading condition or power imbalance makes the power flow control difficult. Moreover, different adaptive droop controllers, which are introduced to alleviate this problem, need a fast communication link between converters [19–22].

2.2. Ratio control

This control method is a flexible droop controller in which power flow in the MTDC network is governed by the modification of droop characteristics, accordingly. However, droop modification needs a communication link that increases the costs and reduces the reliability [25]. In addition, calculation of appropriate ratios to achieve a desired power flow in a large scale MTDC network is a complex process [15].

2.3. Master–slave control

In this control method, only one station, *i.e.* master, regulates DC voltage and the others, called slaves, operate in the power control mode. Outage of the master or reach of its nominal rating may lead to voltage instability in consequence of having no converter to control the network voltage. Due to such problems, this control method is not usually employed in MTDC networks [15,26].

2.4. Priority control

A combination of the master–slave and droop control methods has been introduced in the priority controller. Equipping slave converters with the droop controller gives them the ability of keeping the network stable in a new equilibrium point when the master is unable to control the network voltage [27]. In addition to disadvantages of the droop controller, coordination of this control method with the high number of converters in a large scale MTDC network is a complicated process [15].

2.5. Voltage margin control

This control method is a master–slave controller with additional backup slave converters to regulate the network voltage, if needed. Passing a predefined voltage margin activates the voltage control mode of a backup slave converter without need of any communication link [28]. As one of its drawbacks, only one converter regulates the network voltage at a given time. Also, cascading several PI controllers reduces the dynamic response of this method. In addition, limited range of allowable DC voltage restricts the maximum number of converters that can be controlled by the voltage margin controller.

3. Adaptive time-graded DC voltage control method

The proposed adaptive time-graded control method is based on the fundamental concepts of power system protection. Therefore, these concepts are briefly described and then, the proposed control method is introduced in details.

3.1. Time-graded protection

The selectivity, which means the interruption of smallest area due to fault occurrence, is the basis of each protection scheme. The purpose is isolation of the smallest possible area to continue supplying healthy parts of the network. In addition, interruption of smallest area makes identification of the faulted part easier. However, the speed of a protection scheme is inversely affected by the selectivity. A selective protection scheme can be achieved in different trends, but the most straightforward one is time-graded protection. The idea is grading the operating times of the chain of relays in such a way that the closest relay to the fault spot operates first. In this scheme, each fault triggers two adjacent relays having different operating times. Thus, an inherent backup protection is provided for closest relay to the fault spot. Choosing the proper grading time between these two adjacent relays, is of essential importance to prevent relay-interference. The procedure of calculating proper operating times for the chain of relays is known as relay coordination.

To increase the speed of the time-graded protection scheme for critical areas, an overcurrent relay with inverse time characteristic can be additionally equipped with an instantaneous protection unit. It should be noted that this unit operates almost instant, so it cannot provide a backup protection for its adjacent protected areas. The reclosing action is another concept that is employed in the proposed control method. In a protection scheme, automatic reclosing means reconnecting the interrupted area to bestow the opportunity of continuing the supply after transient faults. Despite the preference of reclosing action with shorter time interval, it mainly depends on the

extinguishing ability of circuit breakers [29].

3.2. Proposed DC voltage control method

The flowchart of the proposed time-graded control method, shown in Fig. 1, consists of three main units that are coincided with the abovementioned protection concepts. In normal conditions, only the slack converter regulates DC voltage of the MTDC network with a simple PI controller. And the others, which are equipped with the proposed controller, operate in the power control mode. Any power imbalance in the MTDC network changes all terminal voltages more or less, and the slack converter, in consequence, tries to tackle this imbalance aim to keep DC voltage inside a predefined range. In case of the slack outage or reach of nominal rating, the control over the network voltage will be missed, resulting continued DC voltage decrease/increase. In such undesirable conditions, the time-graded unit of proposed controller turns a sufficient number of converters to the voltage control mode. For this purpose, each converter continually measures its

terminal voltage as the indicator of power imbalances. Violation of the upper ($V_{cri,u}$) or lower ($V_{cri,l}$) critical voltage limits of i^{th} converter indicates a severe condition that cannot be compensated by the others, so this converter should response immediately. Therefore, this converter instantaneously turns to the voltage control mode, preventing consequences of under or over-voltages. Since only those converters that experience severe voltages turn to the voltage control mode, the others can continue to control the power flow in the MTDC network.

In consequence of a power imbalance in the MTDC network, terminal voltage of each converter firstly trespasses a narrower voltage range with $V_{tri,u}$ and $V_{tri,l}$ as its upper and lower limits, respectively, before reaching to the critical limits. Hitting these narrower limits triggers the time-graded units of some converters and makes them ready to turn their control mode. Preventing unnecessary mode shifting, the mode change delay (MCD_i) is proposed. At first, the highest priority converter with smallest MCD changes its control mode to return terminal voltage inside the range. If it can regulate terminal voltage, other triggered time-graded units do not complete their operation and consequently, stay in the power control mode. Otherwise, the second priority converter with larger MCD , compared to the highest priority one, turns to the voltage control mode after its predefined delay. This mode changing goes on until the minimum number of converters in the voltage control mode can regulate the network voltage. In other words, stable voltage in an n -converter MTDC network after a severe event is the result of operation of m and $n-m$ converters in the voltage and power control modes, respectively.

Ideally, only one converter regulates network voltage and the others operate in the power control mode to govern the power flow in the MTDC network. Therefore, the mode-changed converters should incessantly evaluate the condition and return to the power control mode. For this purpose, the reclosing unit is predicted in the proposed control method. This unit waits for those conditions in which more converters can operate in the power control mode. Power variation is utilized as an index and for variations larger than a predefined value (α), the converter returns to the power control mode. Also, the operation of time-graded unit may lead to large transients in active power of converters. Thus, to prevent inopportune mode-returning due to these transients, a reclosing delay (RD_i), seen in Fig. 1, is considered before mode changing.

After operation of the reclosing unit, the presented flowchart in Fig. 1 starts again. Remaining the terminal voltage inside the range means the opportune mode returning by the reclosing unit, so the converter can operate in the power control mode. In contrast, violation of allowable voltage limits indicates the inopportune mode returning and unsuitability of the present condition for operation of this converter in the power control mode. In consequence, the time-graded unit of this converter turns it again to the voltage control mode. A large power variation, therefore, can bring all mode-changed converters back to the power control mode, some of them can stay in this mode and the others return to the voltage control mode.

4. Discussion

The basis of the proposed control method were described in the previous section. To develop this control method for implementation in the large scale VSC-MTDC networks, some important issues are discussed in the following.

4.1. Instantaneous unit

Coordination of time-graded units in a large scale MTDC network results in large MCD values for low priority converters, reducing their response speed to power imbalances. As a result, terminal voltages may reach to intolerable limits and trigger protective equipment such as DC chopper before the reaction of time-graded units [30]. The proposed instantaneous unit is a simple and effective solution to this problem. It

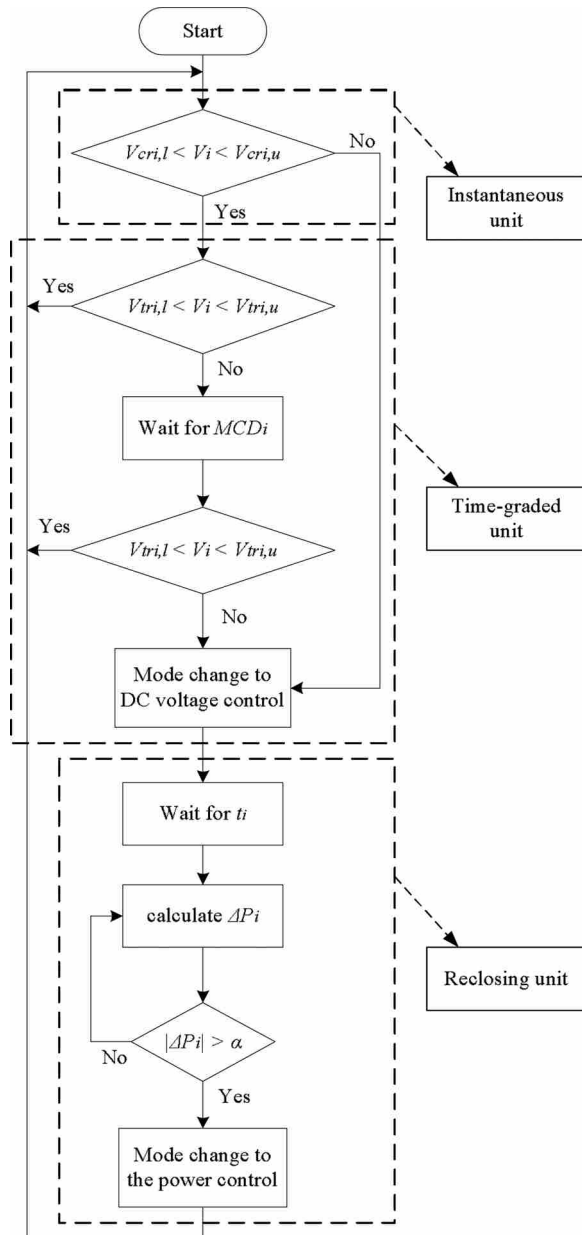


Fig. 1. The flowchart of the proposed time-graded control method.

can independently turn a converter to the voltage control mode, if its predefined limits are violated. As a matter of fact, it is the backup of the time-graded unit to keep the terminal voltage inside the range.

Choosing appropriate critical limits for the instantaneous unit is a crucial decision. Narrower limits may lead to interference between immediate and sequential operation of the instantaneous and time-graded units, respectively. In contrast, protective equipment will undesirably operate before reaction of the instantaneous unit, if wider limits are chosen. To select proper critical limits, some issues should be firstly considered. DC over-voltage is an important issue that is harmful to DC lines insulators, underground cables and converters [30]. Therefore, different ways have been introduced to prevent over-voltage in the MTDC network and diminish its unpleasant consequences [30]. To activate the instantaneous unit of each converter earlier than the protective equipment, its upper critical limit should be chosen slightly smaller than its maximum tolerable voltage. This voltage gap, in addition to a safety margin, covers the inherent delay of mode changing in a practical network.

Decreasing DC voltage below a critical limit is another influential issue that should be considered as well. This leads to high current level in converters operating in the power control mode and transmission line overload. Furthermore, dropping network voltage below a specific critical limit cannot be recovered and consequently, results in network instability. Similar to the upper limit, the lower one is selected slightly bigger than the under-voltage limit, considering the practical delay of mode changing.

Regarding these issues, over- and under-voltage limits of each converter should be firstly calculated by offline studies, which include insulation study and stability analysis of the MTDC network, respectively. Then, the critical limits of each instantaneous unit are determined by considering actual mode change delay of its relevant converter.

To govern the independent operation of instantaneous units in a large scale MTDC network, two or more distinct areas can be specified. These areas are determined by assigning similar critical limits to the converters in one area, which differ from that of others. In this zoning scheme, each severe power imbalance, which could not be tackled by the time-graded units, changes the control mode of converters in the highest priority area that is specified with its narrowest critical limits. If these mode-changed converters could not regulate network voltage, other low-priority areas with wider critical limits will be triggered, respectively. However, due to non-uniformity of terminal voltages in the large scale VSC-MTDC network, neighbor converters inherently form a distinct area. This is correct even with considering similar critical limits for the instantaneous units of all converters.

4.2. Time-graded unit

One of the main drawbacks of each time-graded scheme is an accumulated delay due to its sequential operation. A longer MCD should be assigned to the lower priority converter, compared to the higher priority one, preventing unnecessary mode changes. Thus, implementation of this scheme in a MTDC network with a large number of converters results in unacceptable MCDs for lower priority converters. Although, they may not react timely to keep voltage inside the range, their proposed instantaneous units prevent terminal voltages from trespassing their critical limits. Beside this backup operation, the coordination procedure of time-graded units, similar to that of power system protection, is employed instead of constant MCD. In this procedure, a sufficient MCD is considered for a lower priority converter, giving the higher priority one an adequate time for mode changing. For this purpose, a modified inverse definite minimum time (MIDMT) curve is defined in Fig. 2 for the time-graded unit of each converter. The

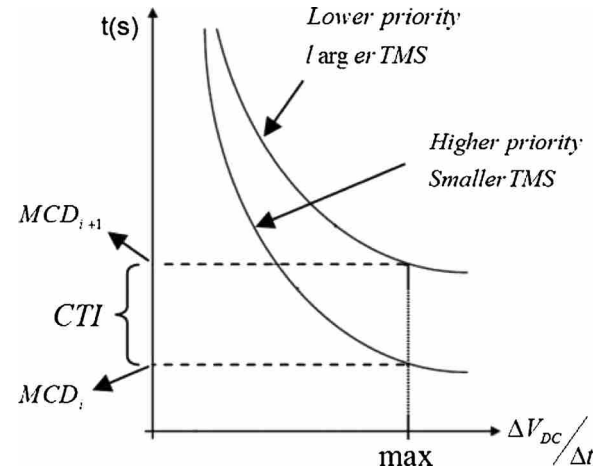


Fig. 2. The modified inverse definite minimum time curve.

inverse dependence of operating time with the rate of voltage change over time can take the severity of the present power imbalance into account. Therefore, a higher rate, which means more severe conditions, uniformly decreases the delay of all time-graded units.

As it is specified in Fig. 2, the operating time of higher priority converter should be less than a coordination time interval (CTI) in which mode change delay as well as a safety margin can be simply considered. Moreover, a time-varying formula such as (1) can be employed to calculate CTI, more effectively:

$$CTI(t) = A_1 t + A_2 \quad (1)$$

where, A_1 relates to the time-dependent delays, whereas A_2 is the representative of constant delays which are identical for all converters. For simplicity, a similar curve is considered for the time-graded units and this curve is only shifted up, as the priority of converter is reduced. One of the best fits for the presented curve in Fig. 2 is (2), which is simply shifted up/down by time multiplier setting (TMS). Thus, TMS specifies the priority of each converter in the sequential action of time-graded units. In addition, to have normal, very or extremely inverse curves, constant coefficients (A , B and α) can be accordingly defined. Therefore, the coordination procedure means calculation of an appropriate TMS value for each converter.

$$MCD_i = \left[\frac{A}{\left(\frac{\Delta V_{DC}}{\Delta t} \right)^\alpha - 1} + B \right] \times TMS_i \quad (2)$$

Without any calculation, zero value is given to the slack converter ($TMS_i = 0$), since it reacts to any voltage change earlier than the others. Next, the operating time of high priority converter should be longer than the slack for all $\Delta V_{DC}/\Delta t$ values. Therefore, maximum probable $\Delta V_{DC}/\Delta t$ of the MTDC network should be obtained by offline simulation of the network under various severe conditions, e.g. converter and transmission line outage, and DC fault. Now, TMS of the high priority converter is calculated such that its operating time for maximum obtained $\Delta V_{DC}/\Delta t$ is more than CTI, compared with that of the slack. The generalized procedure is presented in (3), in which subscript i and $i + 1$ are dedicated to higher and lower priority converters, respectively. Since two similar curves in Fig. 2 tend to converge for large $\Delta V_{DC}/\Delta t$ values, bigger vertical difference than CTI between these two curves for specific $\Delta V_{DC}/\Delta t$ guarantees larger differences for smaller $\Delta V_{DC}/\Delta t$ values. Thus, the calculated TMS values can be confidently employed for all conditions with various $\Delta V_{DC}/\Delta t$.

$$\begin{cases} MCD_i \left(\max \left(\Delta V_{DC} / \Delta t \right) \right) = \left[\frac{A}{\left(\max \left(\Delta V_{DC} / \Delta t \right) \right)^\alpha - 1} + B \right] \times TMS_i \\ MCD_{i+1} \left(\max \left(\Delta V_{DC} / \Delta t \right) \right) \geq MCD_i \left(\max \left(\Delta V_{DC} / \Delta t \right) \right) + CTI \end{cases} \Rightarrow TMS_{i+1}$$

$$= MCD_{i+1} \left(\max \left(\Delta V_{DC} / \Delta t \right) \right) \div \left(\frac{A}{\left(\max \left(\Delta V_{DC} / \Delta t \right) \right)^\alpha - 1} + B \right) \quad (3)$$

Another important issue is selection of the appropriate trigger limits for the time-graded unit. Due to the periodic calculation of optimal power flow (OPF) by power dispatch center (PDC), the last-obtained DC voltage of each converter, with a specific margin e.g. 5%, is selected as its trigger limits. Therefore, those events that cannot be tackled by the slack converter and result in large variations will trigger the time-graded unit. These limits are updated for each planned events in the MTDC network, so there is no malfunction of the time-graded unit in normal conditions. It is worth mentioning that setting these trigger limits requires considering two following notes. First, the limits of each converter should be chosen narrower than the critical limits of its instantaneous unit. Second, due to probable communication failures, some converters may not receive the last updated voltages to correct their trigger limits; therefore, the selected margin should be large enough to prevent malfunction of the time-graded unit.

The impact of mode changing on the AC side network is a serious concern that should be considered in details. More comprehensively, the impact of the proposed control method on the small-signal and transient stability of VSC-MTDC networks [31,32] should be studied in future work. In spite of this fact, a ranking procedure is employed to decrease the unfavorable impacts of mode changing on the stability of adjacent AC networks. As it is introduced in [33], the real-time strength of AC network as well as power margin of the converter can be considered as a capability index to assess the credibility of each converter for DC voltage control. Eq. (4) is an evaluation function to rank the converters at the basis of mentioned indices [33]:

$$Rank_i = \left(\frac{P_r^i - |P_{op}^i|}{P_{max}} \right) \times \left(\frac{SCC_{est}^i}{SCC_{max}} \right) \quad (4)$$

where, P_r^i and P_{op}^i are the rated and present active power of i^{th} converter, respectively and P_{max} is the representative of maximum rated power of all converters in the MTDC network. Correspondingly, SCC_{est}^i is the estimated short circuit capacity (SCC) of AC side grid of i^{th} converter and the maximum SCC of all connected AC grids is indicated by SCC_{max} . The details of estimation procedure can be found in Ref. [33].

Based on this ranking procedure, PDC periodically ranks the converters; selects the highest ranked one as the slack and prioritizes the others at the basis of their calculated ranks. Moreover, TMS values should also be updated, considering new ranks of the converters. For lower ranked converters, which are connected to the weak AC grids, a rate limiter block can be employed in their time-graded units. This block restricts the rate of power variations due to mode changing and consequently, reduces the stability concern of these AC grids.

Another concern about the proposed control method is uncontrolled power exchanges of the mode-changed converters. In other words, working more than one converter in the voltage control mode may be undesirable. Moreover, a severe DC fault can activate the instantaneous or time-graded unit of most converters and consequently, their control mode will be changed. To address this concern, it should be noted that mode changing of most converters means occurrence of a severe event. Such conditions, in the worst scenario, can activate a protection system or even lead to instability of MTDC network. It is obvious that keeping the MTDC network stable in such abnormal conditions is more important than having no control over power flow for a short time. Moreover, the reclosing unit of mode-changed converters continually try to return the control mode. In addition, PDC can also return the control mode of converters, if there is no communication failure.

4.3. Reclosing unit

In a protection scheme, the reclosing action of a circuit breaker is performed after a constant delay, usually with no condition of satisfaction. Similar imitation in the proposed reclosing unit means unconditional returning of a mode-changed converter to the power control mode, after a time interval. As a result, if the abnormal condition continues, the terminal voltage repetitively violates voltage limits. The consequent repetitive mode changes, as a transient phenomenon, may be harmful for the VSC-MTDC network. On the other hand, considering long delays may seem less harmful. The drawback, though, is that the probable opportunities for timely returning of mode-changed converters to the power control mode will be missed. Therefore, the proposed reclosing unit returns the control mode only after a significant power variation. Even though, this may also lead to an unsuccessful mode changing for some converters, the others will operate in the power control mode. Besides, the number of voltage variations is reduced in comparison with the unconditional mode changing.

4.4. A communication-less control method

The proposed control method, indeed, is a local controller that operates only at the base of local measurements. Such an independent relation to the communication link is vital during converter outage or communication failure. Therefore, the proposed control method is an appropriate choice for utilizing in the secondary control level of VSC-MTDC networks. Since the tertiary control level employs a communication link for long-term power flow control, this link can also be utilized by the proposed controller. However, this controller needs no fast communication link to perform its tasks such as selecting trigger limits or ranking procedure. Also, these tasks can be carried out offline in the event of a communication failure.

4.5. Comparison with other DC voltage control methods

An ideal DC voltage control method for the VSC-MTDC network is the one with fast dynamic response, high flexibility to power flow control, high expandability and low communication requirements [34]. Table 1, which summarizes these characteristics for the conventional as

Table 1
Comparing the proposed DC voltage control method with other conventional ones.

Control method	Dynamic response	Expandability	Flexibility	Communication requirement
Droop	High	Medium	Low	Low
Ratio	High	Low	Medium	Medium
Priority	Medium	Medium	Medium	Low
Voltage margin	Low	High	High	Medium
Proposed	Medium	High	High	Low

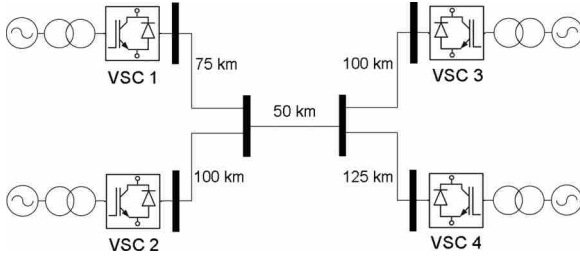


Fig. 3. The structure of the VSC-MTDC test network.

well as the proposed methods, reveals that none of them can satisfy the mentioned indices of an ideal control method. However, this table verifies that the proposed time-graded control method is the best one for employing in the VSC-MTDC network. It is noteworthy that the master-slave control method is not mentioned in Table 1, since it is not appropriate for utilizing in MTDC networks as it was mentioned in Section 2.

5. Simulation results

In this section, the proposed control method as well as conventional droop controller is implemented in a four-terminal VSC-MTDC test network. This network with both control methods is simulated under various conditions by MATLAB/Simulink.

5.1. Four-terminal HVDC test network

Studying a small test system under various conditions can clearly show the main advantages of the proposed control method, comparing with the droop controller. Therefore, a four-terminal MTDC network with presented structure in Fig. 3 is chosen as the test network. The parameters of this network are brought in Table 2, and Table 3 presents the parameters of both proposed and droop controllers.

An average converter model with inner vector control loops in dq -reference frame, which is validated against detailed switched model, is employed in the test network [32]. It is assumed that the converters of test network are prioritized and enumerated by the introduced ranking procedure. For the proposed controller, VSC1 is selected as the slack converter, whereas the others operate in the power control mode with similar control parameters. The active power references of these converters are periodically specified by PDC. For the droop controller, similar gains are assigned to VSC2 and VSC3, while a smaller droop is considered for VSC1.

5.2. Normal conditions

It is assumed that all converters are in-service and operate in a steady state condition. At $t = 0.1$ s, PDC decreases the power reference of VSC4 to 20 MW and after 0.4 s, the reference is changed to -10 MW for delivering power to its connected AC system. The resultant terminal voltages due to operation of the proposed and droop controllers are

Table 2
Parameters of the VSC-MTDC test network.

AC grid	Rated voltage	145 kV
	Rated SCC	10000 MVA
	Rated DC voltage	100 kV
VSC	Rated power of VSC1 to VSC3	100 MW
	Rated power of VSC4	150 MW
	DC Capacitor	150 μ F
	Resistance	0.0133 Ω /km
DC line	Inductance	0.8273 mH/km
	Capacitance	0.0139 μ F/km

Table 3

Parameters of the proposed and droop controllers.

The instantaneous unit	$V_{cris,l}$	85 kV
	$V_{cris,u}$	115 kV
	$V_{trig,l}$	-5%
	$V_{trig,u}$	$+5\%$
The time-graded unit	A, B, α	0.515, 0.114, 0.02
	CTI	0.01 s
	$\text{Max}(\Delta V_{DC}/\Delta t)$	150 kV/s
	$RD_p, i = 1, 2, 3$	0.01 s
The reclosing unit	α	10 MW
	VSC1	5%
Droop gain	VSC2& VSC3	10%

presented in Fig. 4, respectively. Accordingly, Fig. 5 shows the active power variations of converters.

The slack converter, as seen in Fig. 4(a), keeps its terminal voltage around nominal value in different conditions, whereas the voltages of others are varied. Moreover, Fig. 5(a) shows that VSC2 to VSC4 could regulate their power exchange at their commanded set-points. Such excellent power flow control is achieved by the compensating action of the slack converter. It is seen from Fig. 5(a) that VSC1 absorbs extra power of VSC4 between 0.1 to 0.5 s and then, changes to the rectifier mode at 0.5 s to supply power shortage of the network. In contrast, Figs. 4(b) and 5 (b) show the uncontrolled variations of terminal voltage as well as active power of converters, according to power changes of VSC4. VSC1 participates twice in power exchange, while VSC2 and VSC3 operate similarly, considering their droop gains. As it was expected, the droop controller without a fast communication link cannot control the power flow of the MTDC network.

Due to arbitrary chosen of droop gains in Table 3, it should be verified the superiority of proposed control method over any droop gain characteristics. For this purpose, the test network is simulated under various droop gains when VSC4 is supplying constant 100 MW and both VSC2 and VSC3 have similar droops. Fig. 6(a) shows that for large droops, steady state terminal voltages are outside of the acceptable voltage limit (105 kV), while almost constant voltages inside the range are achieved for the small ones. In fact, a small droop gain is equivalent to the operation of converter in the voltage control mode. Despite these appropriate terminal voltages, Fig. 6(b) illustrates the active power of converters are widely changed with variation of droop gains. This can be compared to the ability of proposed method for precise power flow control in the MTDC network.

5.3. Abnormal conditions

Here, *abnormal* means those conditions in which the slack converter cannot solely regulate the network voltage. Therefore, proper reaction of the controller is crucial. In this section, large changes in the power set-points of VSC4 as well as converter outage, among various severe events, are considered as abnormal conditions.

5.3.1. Large changes in the power set-point

Two scenarios for the set-point changes of VSC4 are considered to evaluate both control methods, more comprehensively.

5.3.1.1. First scenario. It is assumed that the MTDC test network initially operates in a normal condition. At $t = 0.1$ s, the set-point of VSC4 is increased by the PDC to 100 MW. Fig. 8(a) shows that the slack converter, which reaches to its nominal rating, cannot supply this large positive imbalance. Consequently, all terminal voltages increase, as seen in Fig. 7(a), and violate upper trigger limits of the proposed controller. Considering Table 3, trigger limits of each converter are chosen $\pm 5\%$ of its expected voltage that is specified by PDC. As a result, the time-graded units of all converters are activated, but solely the control mode of VSC2 is changed after MCD_2 due to its highest

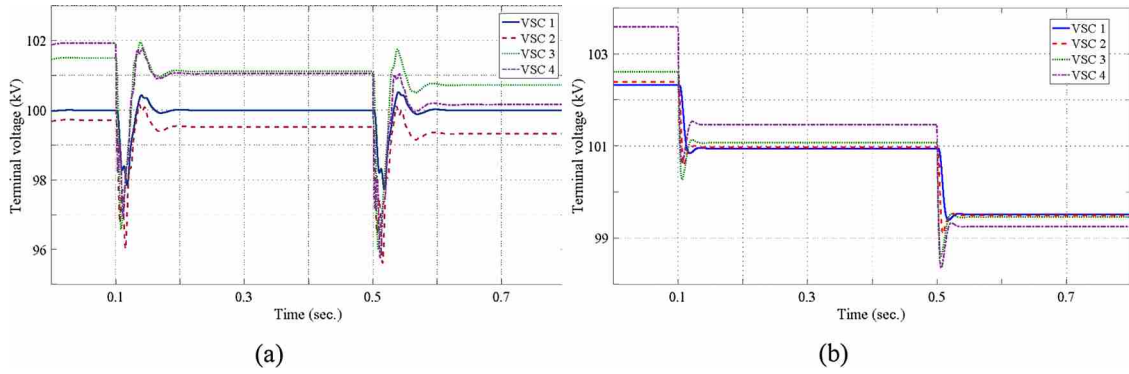


Fig. 4. The terminal voltages in the normal conditions with the (a) proposed and (b) droop controllers.

priority (see Fig. 9). After mode changing of VSC2, Fig. 7(a) shows that terminal voltages are returned inside the range. Now, both VSC1 and VSC2 are slack converters that cooperatively regulate the network voltage.

The power set-point of VSC4 is reduced to 95 MW at $t = 0.5$ s. Fig. 8(a) shows that both VSC1 and VSC2 cover this power imbalance without any more mode changes of others. However, this power variation is not large enough to activate the reclosing unit of VSC2. The next event is a significant change in the power set-point of VSC4, which is commanded by PDC at $t = 0.7$ s. The resultant power variation is large enough to trigger the reclosing unit of VSC2. Returning VSC2 to the power control mode after RD_2 improves power flow control in the MTDC network. In addition, Fig. 8(a) shows that both VSC3 and VSC4 continually follow their power set-points, even during the abnormal condition. Having the maximum possible control over the power flow, presents the outstanding features of the proposed controller.

In contrast, there is the droop control method in which VSC2 to VSC4 cooperatively react to each demanded power of VSC4, as it is seen in Fig. 8(b). Similar to previous results, terminal voltages are simply changed according to power variations of VSC4. Oppositely, Figs. 7(a) and 8 (a) show that VSC1 and VSC2 regulate network voltage in the abnormal condition, while VSC3 and VSC4 exchange their predefined powers. Moreover, VSC2 returns to the power control mode. These results verify the superiority of the proposed controller in such abnormal conditions.

5.3.1.2. Second scenario. Since no new result can be obtained by considering the droop controller in other abnormal conditions, only the proposed controller is studied hereafter to prevent excessive figures. The initial conditions are assumed similar to the first scenario; the first change in the power set-point of VSC4 is occurred at $t = 0.1$ s that may be commanded by mistake. The consequent rising voltages reveal inability of the slack converter to supply this power imbalance. Figs. 10 and 11 show that turning VSC2 to the voltage control mode

after MCD_2 is insufficient, since it reaches again to its nominal rating.

As the next priority converter, VSC3 reacts to this imbalance after a longer delay (MCD_3), which can be clearly seen in Fig. 12 as well as the zoomed voltages in Fig. 10. As a result, these three converters could cooperatively return terminal voltages within the range. Comparing Figs. 9 and 12 reveals the smaller value of MCD_2 in the second scenario, comparing with the first one, due to adaptivity of the proposed control method.

At $t = 0.5$ s, the power set-point of VSC4 is changed (or corrected) to 80 MW that leads to a large power imbalance in the network. Therefore, the reclosing units of both VSC2 and VSC3 return their relevant converters to the power control mode after RD_2 and RD_3 , respectively. The rising voltages in Fig. 10 mean that this new condition is not appropriate for changing the control mode of both converters. Consequently, the time-graded unit of VSC2, as a higher priority converter, turns it again to the voltage control mode after MCD_2 . The resultant voltages are acceptable, so the triggered time-graded unit of VSC3 is not activated, due to larger MCD_3 in comparison with MCD_2 . Therefore, VSC3 remains in the power control mode and the network voltage is regulated by the cooperation of VSC1 and VSC2.

From power flow viewpoint, operation of VSC2 in the power control mode is preferred. In accordance, the reclosing unit of VSC2 waits for another large power imbalance to return the control mode. However, due to periodic updates of power set-points by PDC, when there is no communication failure, VSC2 will finally return to the power control mode. In addition, PDC can also directly change the control mode of VSC2, if needed.

5.3.2. Converter outage

Converter outage due to its AC/DC sides or internal faults leads to disability of the converter to exchange power or regulate the network voltage. If the converter cannot come back into the service, PDC will manage the power flow in the network without faulted converter. Nevertheless, the secondary control level is responsible for immediate

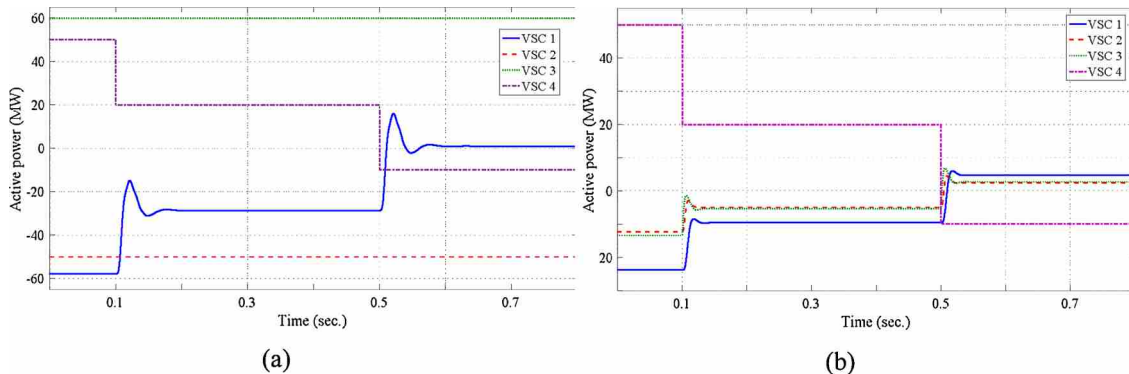


Fig. 5. The active power variations in the normal conditions: (a) proposed and (b) droop controllers.

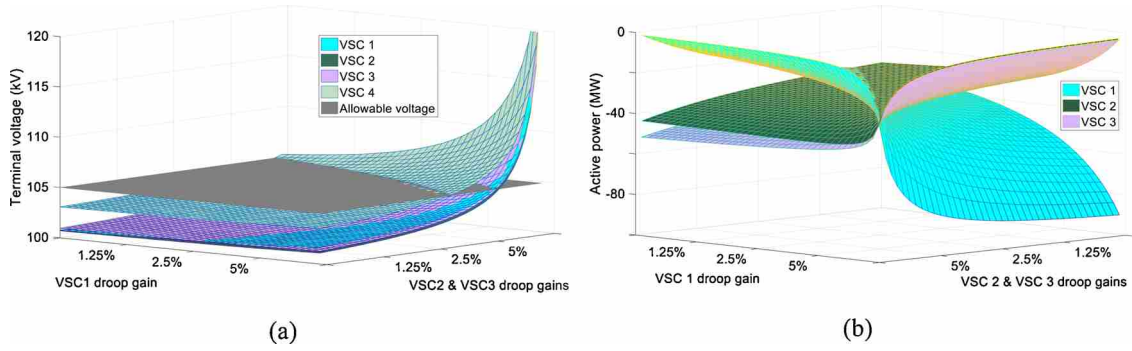


Fig. 6. (a) The terminal voltages and (b) active power exchanges for various droops.

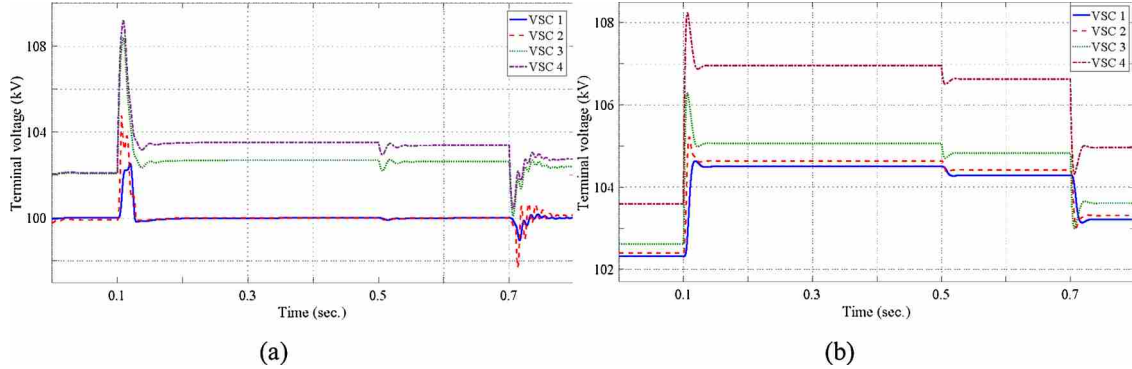


Fig. 7. The terminal voltages in the first scenario with the (a) proposed and (b) droop controllers.

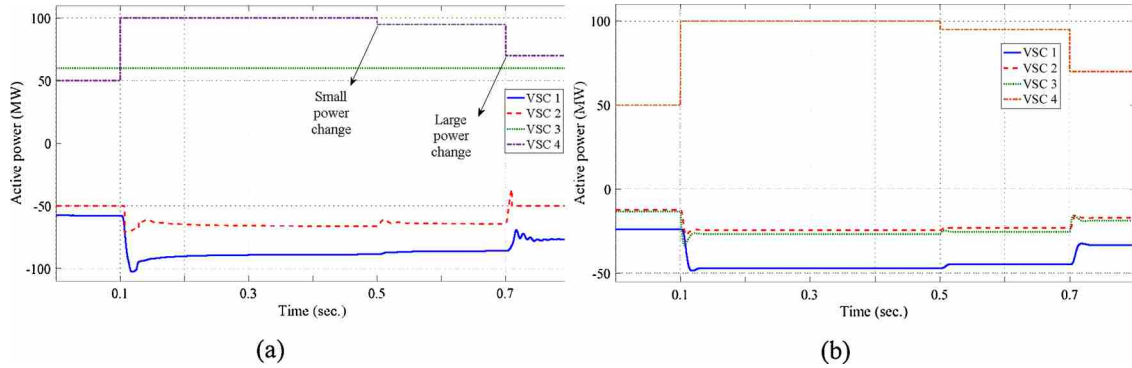


Fig. 8. The active power variations in the first scenario with the (a) proposed and (b) droop controllers.

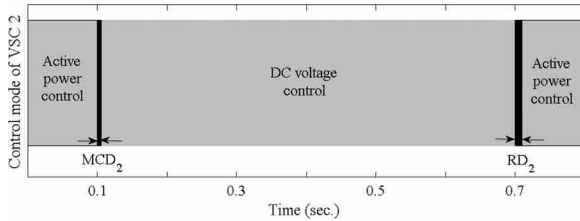


Fig. 9. The control modes of VSC2 in the first scenario.

reaction to keep the MTDC network stable. Initially, VSC1 is regulating network voltage and the others are operating in the constant power mode. Suddenly, an unplanned outage of VSC2 occurs at $t = 0.1$ s that lasts for 0.4 second. The slack converter tries to regulate DC voltage, but Fig. 13 shows that it could not supply the missed power of VSC2 due to reach of nominal rating. The consequent rising voltage triggers the time-graded unit of the other two, but VSC3 with smaller MCD turns to the power control mode, earlier (see Fig. 14). Fig. 13 shows that this proper mode changing returns terminal voltages inside the range.

Recovering VSC2 at $t = 0.5$ s results in a large power imbalance that activates the reclosing unit of VSC3 and consequently, this converter returns to the power control mode after RD₃. Accordingly, returning the MTDC network to its pre-outage condition can be seen in Fig. 13.

To compare the proposed and droop control methods more precisely, deviation of the obtained steady state results for various conditions are tabulated in Table 4. It should be noted that “V” and “P” in this table are representative of voltage and power control modes, respectively. Moreover, deviation of power exchange from its desired value cannot be defined for VSC1 in the proposed controller, since it operates as the slack converter. To obtain the per-unit value of voltage deviations, 100 kV is chosen as the base voltage, and deviation of active power exchange in each time interval is defined by (5):

$$\Delta P_i = \frac{P_{op}^i - P_{ref}^i}{P_{ref}^i}, \quad i = 2, 3 \quad (5)$$

where, P_{ref}^i is the active power reference of i^{th} converter. Since the negative or positive sign of both P_{ref}^i and P_{op}^i is considered in (5), $\Delta P_i > -100\%$ in Table 4 indicates the change of power flow direction in i^{th}

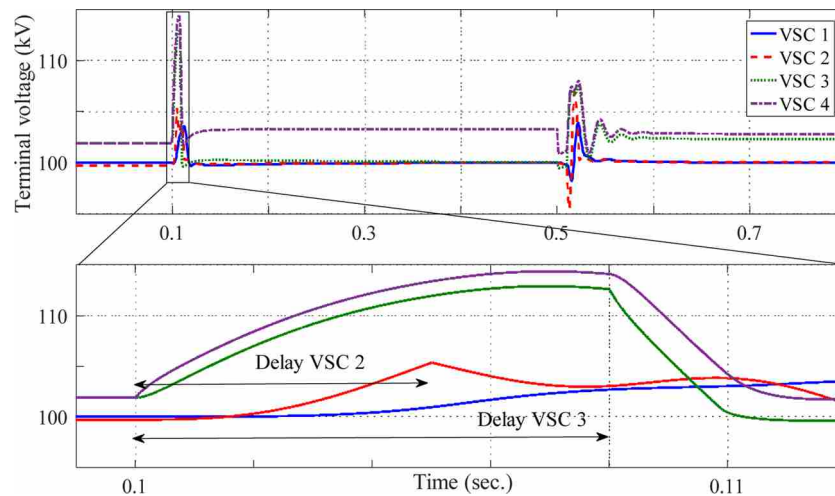


Fig. 10. The terminal voltage of converters for large changes in the power set-points (second scenario) with zoomed voltages around $t = 0.1$ s.

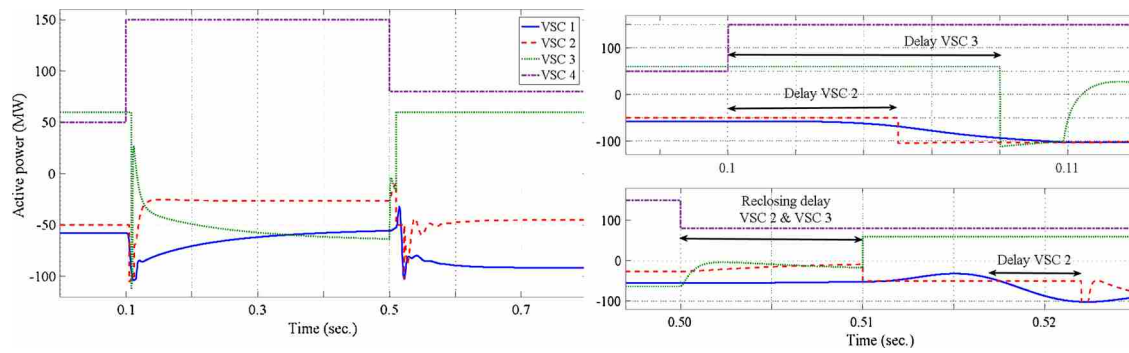


Fig. 11. The active power variations for large changes in the power set-points (second scenario) with zoomed powers (right side) around $t = 0.1$ s and 0.5 s.

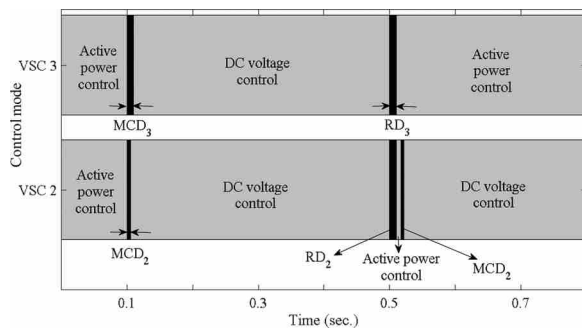


Fig. 12. The control modes of VSC2 and VSC3 in the second scenario.

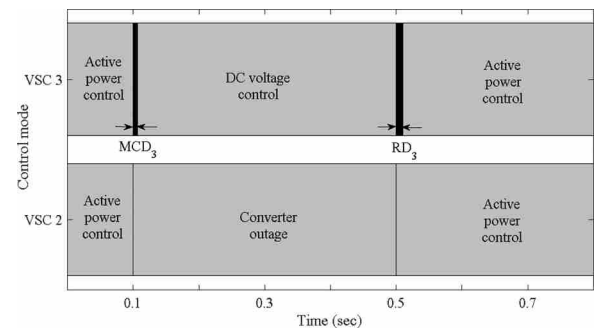


Fig. 14. The control modes of VSC2 and VSC3 in the converter outage condition.

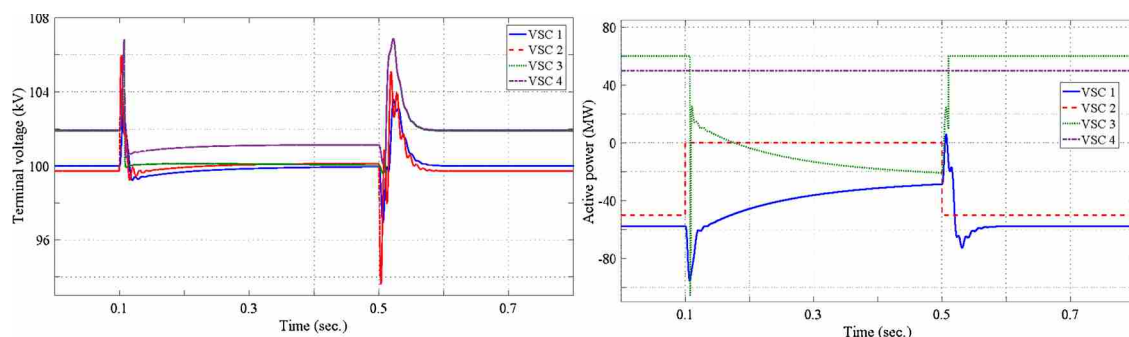


Fig. 13. The terminal voltages and active powers in the converter outage condition.

Table 4
Deviation of the obtained steady state numerical results from their desired values.

Deviation from nominal values (%)	Normal condition (small changes in the power set-point of VSC4)										Abnormal condition (large changes in the power set-point of VSC4)										Outage of VSC2		
	Proposed					Droop					Proposed					Droop					Proposed		
	Before	Small decrement	Change to inverter	Before	Change to inverter	Before	Small decrement	Large increment	Small decrement	Before	Before	Small decrement	Large increment	Small decrement	Before	Before	Small decrement	Large increment	Small decrement	Before	Before	Occurrence	Recover
	Before	Small decrement	Change to inverter	Before	Change to inverter	Before	Small decrement	Large increment	Small decrement	Before	Before	Small decrement	Large increment	Small decrement	Before	Before	Small decrement	Large increment	Small decrement	Before	Before	Occurrence	Recover
	Before	Small decrement	Change to inverter	Before	Change to inverter	Before	Small decrement	Large increment	Small decrement	Before	Before	Small decrement	Large increment	Small decrement	Before	Before	Small decrement	Large increment	Small decrement	Before	Before	Occurrence	Recover
VSC1	ΔV_1	0	-0.48	2.3	0.9	-0.48	0	0	0	2.3	2.3	4.3	4.5	4.3	3.2	0	0	0	0	0	0	0	0
VSC2	Mode	V	-0.28	2.4	1	-0.5	0	33	0	2.4	2.4	4.4	4.6	4.4	3.3	V	0	-47	0	-0.28	0	0	-0.28
VSC3	ΔV_2	-0.48	-0.67	-76	-90	-105	0	28.6	0	0.1	0	-54	-51	-54	-66	0	0	0	0	0	0	0	0
VSC4	Mode	P	0	0.7	1.1	-55	0	2.7	0	P	P	5.1	7	4.8	3.6	P	0	-206	0	1.9	1.9	0	1.9
VSC4	ΔV_3	1.5	0.7	2.6	-73	-105	0	3.5	3.4	2.6	2.4	-144	100	-142	-131	0	0	0	0	0	0	0	0
VSC4	Mode	P	0	0.2	1.5	-75	0	100	95	-122	-122	6.6	100	95	5	P	0	150	80	1.9	1.9	1.1	1.9
VSC4	ΔV_4	1.9	0.2	3.6	20	-10	0	100	95	3.6	3.6	6.6	100	95	70	50	50	150	80	1.9	1.9	1.1	1.9
VSC4	P_{ref}	50	-10	50	20	-10	0	100	95	50	50	95	100	95	70	50	50	150	80	1.9	1.9	1.1	1.9
VSC4	Mode	P	0	0.2	1.5	-75	0	100	95	3.6	3.6	6.6	100	95	5	P	0	150	80	1.9	1.9	1.1	1.9

converter, regarding its reference value.

Table 4 clearly shows that the proposed control method can keep terminal voltages inside the allowable range ($\pm 5\%$), while the droop controller may lead to undesired terminal voltages, which are highlighted in the table. Moreover, it can be seen that the proposed control method, in contrast with the droop one, can return active power exchange of VSC2 and VSC3 to their reference values in the worse conditions.

6. Conclusion

This paper proposes an adaptive time-graded control method for the power flow and voltage control of VSC-MTDC networks. In normal conditions, the highest priority converter regulates the network voltage and the others control the power flow in the network. Upon happening an undesirable event such as converter outage or large power change, the proposed controller sequentially turns a sufficient number of higher priority converters to the voltage control mode. Therefore, the network voltage is regulated by the minimum number of converters and consequently, the others can be employed for power flow control. Moreover, it is seen that the proposed controller continuously evaluates the conditions for returning the mode-changed converters to the power control mode. Therefore, the proposed method can appropriately control the power flow and network voltage, as main challenges in the operation of VSC-MTDC network, under various conditions. Besides, proposing an adaptive procedure to calculate MCD of each converter is another contribution of the present study. Moreover, no need to fast communication link is another remarkable feature of the proposed controller that significantly increases the reliability. The acclaimed outstanding abilities of proposed controller are verified by simulation of a test MTDC network in MATLAB/Simulink under various conditions with both proposed and conventional droop methods. It should be noted that the present study is a first step to introduce a new DC voltage control method for MTDC network, so lots of work still remain toward employing it as a practical controller in actual networks. Beside those issues that are discussed more or less as a guide map, other concerns also should be considered more comprehensively in future work.

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